

SHORT BIO. My research focuses on causal decision making (NeurIPS-25, Preprint, ICLR-26) and causal effect identification (NeurIPS-24). I investigate how to formally model diverse real-world problems within a causal framework and derive optimal decisions from such structured representations. In particular, I develop methods for robust decision-making in settings under environmental uncertainty (NeurIPS-25), domain shifts between training and deployment environments (Preprint), and even scenarios involving counterfactual actions (ICLR-26).

EDUCATIONS

Graduate School of Data Science, Seoul National University Seoul, Republic of Korea
Integrated M.S. and Ph.D. in Data Science Sep 2022 – Feb 2027

Advisor: Prof. Sanghack Lee.

Dissertation: “Generalizing Causal Decision-Making: Markov Equivalence, Transportability and Counterfactuals”.

Overall GPA: 4.02 / 4.3; Ph.D. Qualification Exam partially waived due to outstanding academic performance.

Department of Mathematics and Statistics, Sejong University Seoul, Republic of Korea
B.S. in Applied statistics and Mathematics (Double major), Cum Laude Mar 2016 – Aug 2022

Advisors: Prof. Ji Eun Lee and Jin Hee Yoon.

(2.5 years military service)

Overall GPA: 4.04/4.5; Major GPA: 4.46 (Applied Statistics), 4.38 (Mathematics).

PUBLICATIONS

* for joint first authorship and † for corresponding author.

- ◇ Min Woo Park and Sanghack Lee†. [Counterfactual Structural Causal Bandits](#). *International Conference on Learning Representations (ICLR)*, 2026.
- ◇ Min Woo Park, Andy Arditi, Elias Bareinboim†, Sanghack Lee†. [Structural Causal Bandits under Markov Equivalence](#). *Neural Information Processing Systems (NeurIPS)*, 2025, Columbia CausalAI Laboratory, Technical Report (R-122)
- ◇ Min Woo Park and Sanghack Lee†. [On Transportability for Structural Causal Bandits](#). *arXiv preprint*, 2025. under review.
- ◇ Yonghan Jung†, Min Woo Park, and Sanghack Lee†. [Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference](#). *Neural Information Processing Systems (NeurIPS)*, 2024.
- ◇ Min Woo Park*, and Ji Eun Lee*†. [Computation of the iterated Aluthge, Duggal, and Mean transforms](#). *Filomat*, 2023.
- ◇ Min Woo Park*, Taehui Yun*, Youngin Jang, Younsuk Yeom, Jonghwan Kim, LG AI Research, and Sanghack Lee†. [Breaking Bad: Component-wise Parent Deletion for Score-Based Causal Discovery](#). under review.
- ◇ Yesong Choe, Yeahoon Kwon, Min Woo Park, and Sanghack Lee†. [Canonical Domain Reduction for Partial Counterfactual Identification](#). under review.
- ◇ Ji Eun Lee*† and Min Woo Park*. [Convergence of the iterated mean transforms of a \$2 \times 2\$ matrix](#). under review.

AWARDS AND HONORS

Graduate Student Instructor (GSI) scholarship | Seoul National University Spring 2024, Spring 2025, Spring 2026

BK21 scholarship Fall 2024, Fall 2025

First place merit scholarship | Sejong University Fall 2020, Fall 2021

Fourth place award | The 9th College of Natural Sciences Academic Conference, Sejong University Nov 2021

▷ Mitigating spurious regression in time-series data via Granger causality.

Third place award | The 8th College of Natural Sciences Academic Conference, Sejong University Nov 2020
 ▷ The significance of the Euler’s number and its applications.

PROJECTS

Towards More Scalable Score-based Causal Discovery | LG AI Research Sep 2025 – Aug 2026

- ▷ Leading a project on scalable score-based causal discovery algorithms.
 (First-author submission under review.)
 - Developed a novel operator enabling score-based causal discovery algorithms to escape local optima.
 - Investigating parallelization strategies for score-based causal discovery algorithms.

Hyundai NGV Data Boot-Campus 2025 | Hyundai Motor July 2025 – Sep 2025

- ▷ Developed high-voltage battery performance prediction technology.

Hyundai NGV Data Boot-Campus 2024 | Hyundai Motor July 2024 – Sep 2024

- ▷ Developed an intelligent system for predicting vulnerable regions of automotive body panel stiffness using differential geometrybased automatic curvature analysis.

Deep Reinforcement Learning for Recommendation Systems | Coursework Project Mar 2024 – Jun 2024

- ▷ Developed a recommendation system based on DDPG, leveraging multi-modal item features (video, audio, and text) to improve recommendation performance.
 - Source code available at github.com/MinwooPark96/ddpgRLRS.

Optimal Mixed Policies Completeness | Oxford University Aug 2023 – Feb 2024

- ▷ Studied complete characterization of conditions under which the intervention effect of a variable is positive.
 Acknowledged for helpful comments in [Toward a Complete Criterion for Value of Information in Insoluble Decision Problems](#). *Transactions on Machine Learning Research* (TMLR), 2024.

TEACHING ASSISTENTS

Causal Inference for Data Science Spring 2024, Fall 2024, Spring 2025, Spring 2026

- ▷ Topics included Bayesian networks and graphical causal inference.

Machine Learning and Deep Learning for Data Science II Fall 2023

- ▷ Covered kernel methods, Gaussian processes, and related topics in machine learning.

Basic Mathematics for Artificial Intelligence (Undergraduate) Spring 2022

- ▷ Covered linear algebra, basic statistics, and probability theory.

ACADEMIC SERVICES, PROFESSIONAL TALKS AND POSTERS

Counterfactual Structural Causal Bandits

ICLR 2026 poster session | Rio de Janeiro, Brazil Apr 2026 (scheduled)

Structural Causal Bandits under Markov Equivalence

ExploreCSR poster session | Seoul, Republic of Korea Dec 2025

NeurIPS 2025 poster session | San Diego, USA Dec 2025

JKAIA 2025, NeurIPS preview session | Seoul, Republic of Korea Nov 2025

SNU GSDS student research seminar | Seoul, Republic of Korea Apr 2025

Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference

NeurIPS 2024 poster session | Vancouver, Canada Dec 2024

SNU GSDS student research seminar | Seoul, Republic of Korea Dec 2024

Conference Reviewer: 2026 NeurIPS, UAI, ICLR, ICML, 2025 ICML

REFERENCE

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Jin Hee Yoon | Dept. of AI and Information Technology, Sejong University
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